
DISEASES AFFECTING THE STRUCTURE OF BONE

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Bone Cell Coordination and Disease Mechanism

Bone Remodeling: Osteoblasts, Osteocytes, Osteoclasts

Annual Turnover: 5–10% of adult bone replaced yearly

Disease Impact:

- Imbalance in calcium/phosphate levels
- Dysregulation of bone cells
- Abnormal bone and tooth formation

Bone Dysplasias and Imaging Modalities

Bone Dysplasias:

- Fibro-osseous lesions: fibrous tissue replaces normal bone
- Poorly formed or immature bone

Imaging Techniques:

- Intraoral: High resolution, detects subtle jaw changes
- Panoramic/Skull: Extent of bone involvement
- CBCT/MDCT: 3D visualization
- Nuclear Medicine: Bone metabolism

Radiologic and Dental Manifestations

Imaging Changes:

- Altered bone size, shape, trabeculae, marrow spaces
- Cortical thinning/thickening
- Increased or decreased bone density

Dental Effects:

- Delayed/accelerated eruption
- Hypoplasia, hypocalcification
- Loss of lamina dura

Changes in Bone Observed in Systemic Disease^a

	BONES				
			TRABECULAE		
Systemic Disease	Density	Size of Jaws	Increase	Decrease	Granular
Hyperparathyroidism	Decrease	No	Yes	Yes	Yes
Hypoparathyroidism	Rare increase	No	No	No	No
Hyperpituitarism	No	Large	No	No	No
Hypopituitarism	No	Small	No	No	No
Hyperthyroidism	Decrease	No	No	No	No
Hypothyroidism	No	Small	No	No	No
Cushing syndrome	Decrease	No	No	Yes	Yes
Osteoporosis	Decrease	No	No	Yes	No
Rickets	Decrease	No	No	Yes	No
Osteomalacia	Rare decrease	No	No	Rare decrease	No
Hypophosphatasia	Decrease	No	No	Yes	No
Renal osteodystrophy	Decrease; rare increase	Large	Rare	Yes	Yes
Hypophosphatemia	Decrease	No	No	Yes	Yes
Osteopetrosis	Increase	Large			
Sickle cell anemia	Decrease	Large maxilla			
Thalassemia	Decrease	Large maxilla			

Effects on Teeth and Associated Structures^a

Systemic Disease	Hypocalcification	Hypoplasia	Large Pulp Chamber	Loss of Lamina Dura	Loss of Teeth	Eruption
Hyperparathyroidism	No	No	No	Yes	Rare	No
Hypoparathyroidism	No	Yes	No	No	No	Delayed
Hyperpituitarism	No	No	No	No	No	Supraeruption
Hypopituitarism	No	No	No	No	No	Delayed
Hyperthyroidism	No	No	No	Thin	Yes	Early
Hypothyroidism	No	No	No	No	Yes	Delayed
Cushing syndrome	No	No	No	Partial	No	Premature
Osteoporosis	No	No	No	Thin	No	No
Rickets	Yes, enamel	Yes, enamel	No	Thin	No	Delayed
Osteomalacia	No	No	No	Thin	No	No
Hypophosphatasia	Yes	Yes	Yes	Yes	Yes	Delayed
Renal osteodystrophy	Yes	Yes	No	Yes	No	No
Hypophosphatemia	Yes	Yes	Yes	Yes	Yes	No
Osteopetrosis	Yes	Rare	No	Thick	Yes	Delayed

Metabolic Bone Abnormalities – Osteopenia

Disease Mechanism:

- Osteopenia reflects increased bone resorption.
- Radiologic signs: generalized loss of cancellous architecture and cortical density.
- Bone becomes more radiolucent.
- Bone mass peaks around age 30, then declines (8%/decade in females, 3% in males).

Clinical Features:

- Often asymptomatic early on.
- Diagnosed via DEXA scan; may progress to osteoporosis.
- Common in postmenopausal women.
- Risk of fractures (distal radius, femur, ribs, vertebrae).
- Bone pain may occur.

Metabolic Bone Abnormalities – Osteopenia

Imaging Features:

- Generalized reduction in bone density.
- Teeth appear more radiopaque against radiolucent bone.
- Trabeculae: shorter, more delicate.
- Cortical thinning, especially in mandible.
- Lamina dura may be thinner.
- Panoramic/periapical imaging not reliable for diagnosis.

Management:

- Lifestyle: weight-bearing exercise.
- Supplements: calcium, vitamin D, estrogen.
- Medications: bisphosphonates, RANKL inhibitors.



Rickets and Osteomalacia – Disease Overview

Disease Mechanism:

Rickets: Affects children; soft, fragile bones due to vitamin D, calcium, or phosphate deficiency.

Osteomalacia: Adult form of rickets; similar nutritional or metabolic causes.

Causes:

- Inadequate dietary intake or sun exposure
- Malabsorption or synthesis issues (liver/kidney)
- **Hypophosphatemic Rickets:** X-linked disorder; renal phosphate wasting

Effect: Poor calcification → soft bones, abnormal trabeculae, dental defects

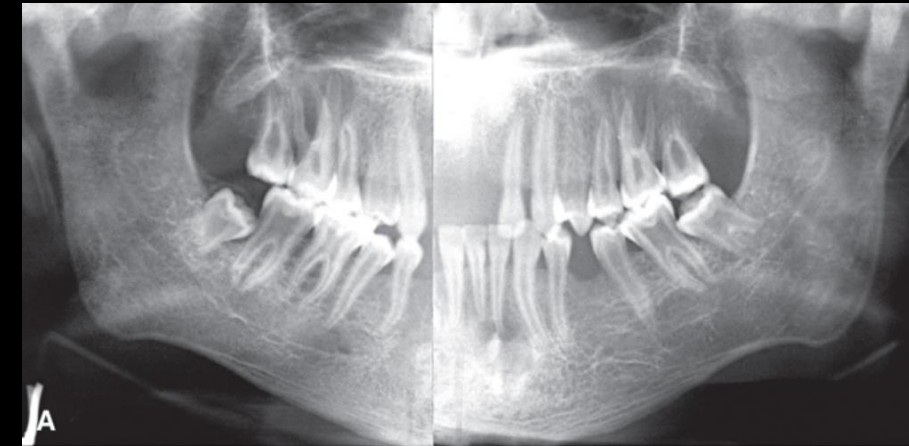
Clinical Features:

Rickets:

- Tetany, craniotabes, bowed legs, short stature
- Delayed tooth eruption, enamel/dentin hypocalcification

Osteomalacia:

- Bone pain, muscle weakness, waddling gait, tetany



Rickets and Osteomalacia – Disease Overview

Imaging Features:

Teeth & Jaws:

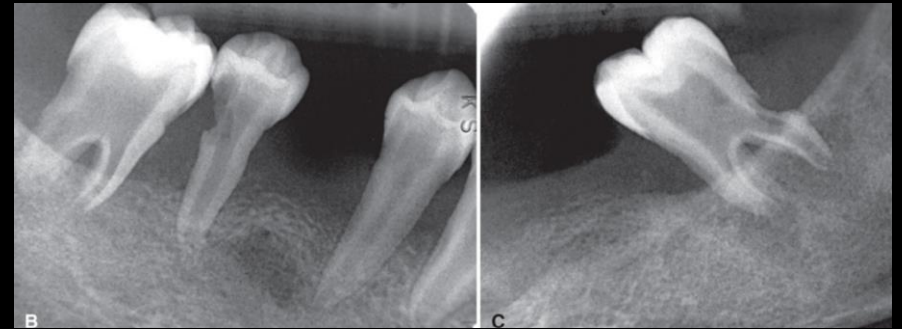
- Thin enamel, large pulp chambers/root canals
- Delayed eruption, thin lamina dura, premature tooth loss

Skeleton:

- Widened epiphyses, cupping/fraying of metaphyses
- Bowing of long bones, greenstick fractures (children), pseudofractures (adults)

Management:

- Vitamin D, calcium, phosphate supplements
- Monitor serum calcium to avoid hypercalcemia
- Long-term risk: nephrocalcinosis



Hypophosphatasia

Mechanism:

- Genetic disorder; reduced alkaline phosphatase activity.
- Leads to accumulation of pyrophosphate → inhibits bone and tooth mineralization.

Clinical Forms:

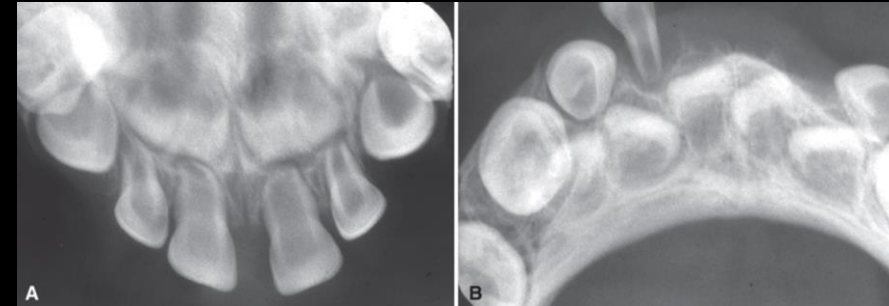
- Six types: 2 perinatal (lethal/benign), infantile, childhood, adult, odontohypophosphatasia.
- Severe forms: early death; mild forms: poor growth, fractures, premature tooth loss.

Dental & Skeletal Features:

- Teeth: thin enamel, large pulp chambers, early loss.
- Jaws: osteopenia, thin cortical bone, deficient alveolar process.
- Skeleton: poorly calcified skull, bowed limbs, “beaten copper” skull in older children.

Management:

- No cure; enzyme replacement therapy shows promise in severe pediatric cases.



Osteopetrosis

Mechanism:

- Genetic defect in osteoclasts → impaired bone resorption.
- Results in **dense, brittle bones** prone to fracture and infection.
- **Types:**
- **Recessive (congenita):** severe, early onset, often fatal.
- **Dominant (tarda):** milder, may be asymptomatic.

Clinical Features:

- Fractures, anemia, nerve compression (blindness, deafness).
- Dental issues: delayed eruption, malformed teeth, early loss.
- Prone to **osteomyelitis**, especially in the mandible.

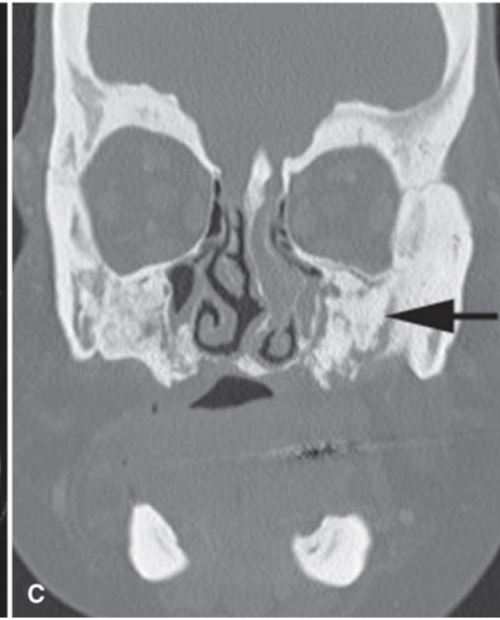
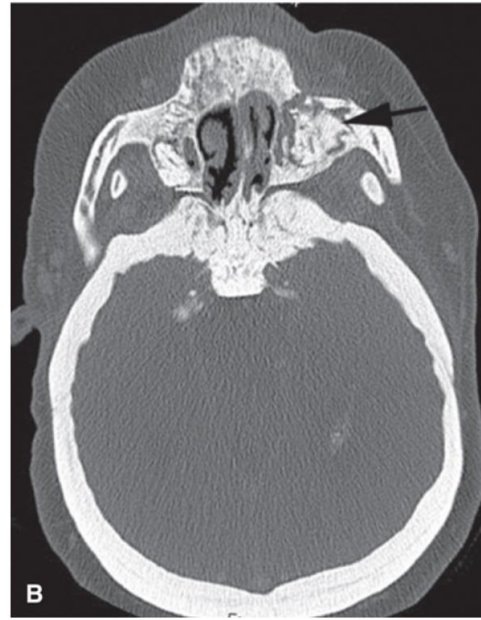
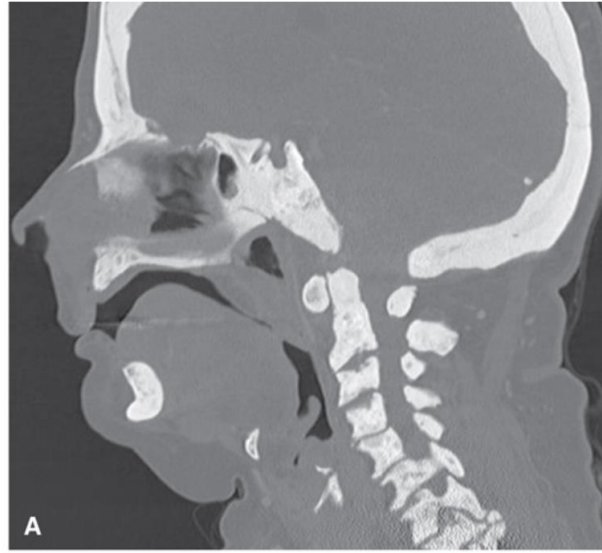
Imaging:

- **High bone radiopacity**, loss of trabecular detail.
- Thick lamina dura, indistinct root outlines.
- Prominent inferior alveolar canal.

Management:

- **Bone marrow transplant** to restore osteoclast function.
- Steroids for blood issues; antibiotics + hyperbaric oxygen for infections.





Systemic Sclerosis

Mechanism & Clinical Features:

- Connective tissue disease → excess collagen → tissue hardening.
- Localized (skin/teeth) or systemic (organs).
- Common in ages 30–50; females 3× more affected.
- Skin: tight, leathery; limited jaw opening.
- Oral: xerostomia, caries, periodontal disease.
- Systemic: heart, lung, kidney complications.

Imaging & Management:

- Nonconcentric PDL space widening (localized/generalized); lamina dura intact.
- Mandibular erosions at muscle attachment sites (angles, coronoid, condyles).
- Differential: trauma, orthodontics, malignancy.
- Management: periodic panoramic imaging to monitor mandibular integrity and prevent fractures.



Bone Dysplasias

Mechanism:

- Bone dysplasias involve replacement of normal bone with immature bone and fibrous tissue.
- Caused by dysregulated osteoblast/osteoclast activity.
- Not premalignant or neoplastic.

Cemento-Osseous Dysplasia (COD):

- Abnormal bone metabolism → fibrous tissue + immature bone.
- **Types:**
 - **PCOD:** localized to one sextant.
 - **FCOD:** affects multiple sextants; may be familial.
- Risk of infection increases as lesions mature and vascularity decreases.

Clinical Features:

- Common in women, especially non-Caucasian, in 4th–5th decades.
- Usually asymptomatic; may cause swelling or low-grade pain.
- FCOD may lead to osteomyelitis if infected.

Imaging Features & Management

Imaging Features:

- **Location:** Periapical areas, often above the inferior alveolar canal.
-

Stages:

- Early: radiolucent, loss of lamina dura.
- Intermediate: mixed radiolucent/radiopaque, cotton-wool or swirling pattern.
- Mature: fully radiopaque with thin radiolucent rim.
- **FCOD:** Multifocal, centered in alveolar processes.

Effects on Adjacent Teeth:

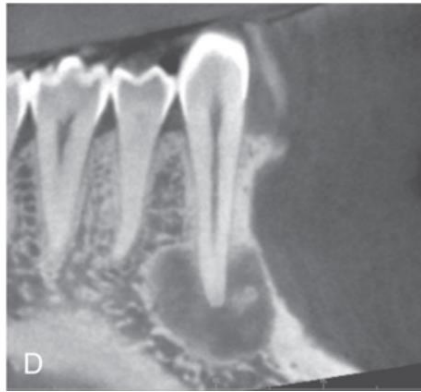
- Loss of lamina dura, widened/indistinct PDL space.
- Tooth structure unaffected; hypercementosis common in FCOD → difficult extractions.

Differential Diagnosis:

- Rarefying osteitis, cementoblastoma, odontome, dense bone island, cemento-ossifying fibroma, Paget disease, osteomyelitis.

Management:

- No treatment needed; panoramic imaging for extent.
- Emphasize oral hygiene to prevent infection.
- Avoid biopsy due to risk of secondary infection.
- If exposed under dentures → risk of osteomyelitis; treat with antibiotics + hyperbaric oxygen if needed.





Fibrous Dysplasia

Mechanism:

- Postzygotic mutation of **GNAS1 gene** → abnormal bone remodeling.
- Cancellous bone replaced by fibrous tissue + immature woven bone.
- Trabeculae: short, irregular, randomly oriented, increased in number.

Clinical Forms:

- **Monostotic (70%)**: single bone, often jaws.
- **Polyostotic**: multiple bones; may include:
 - **Jaffe Syndrome**: + café au lait spots.
 - **McCune–Albright Syndrome**: + endocrine dysfunction.

Clinical Features:

- **Unilateral bone involvement.**
- **Facial swelling, jaw deformity.**
- **Rare: pain, fractures, neural deficits (if skull base involved).**

Fibrous Dysplasia

Imaging Features:

- **Location:** Maxilla > Mandible; often posterior.
- **Periphery:** Poorly defined, “blending” transition zone.

Internal Patterns:

- Ground glass, peau d’orange, fingerprint, cotton-like.
- Radiolucent early → mixed/radiopaque mature.

Effects:

- Bone enlargement (fusiform in mandible).
- Displacement of teeth, sinus walls, and canals.
- Lamina dura indistinct; PDL space narrowed.



Fibrous Dysplasia

Differential Diagnosis:

Similar Patterns:

- Healing bone, PCOD (older, multifocal), simple bone cavity.

Metabolic Diseases:

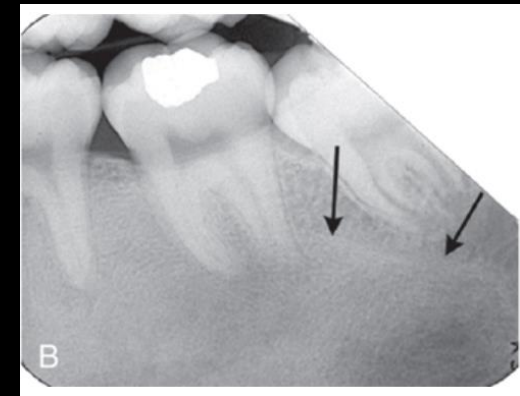
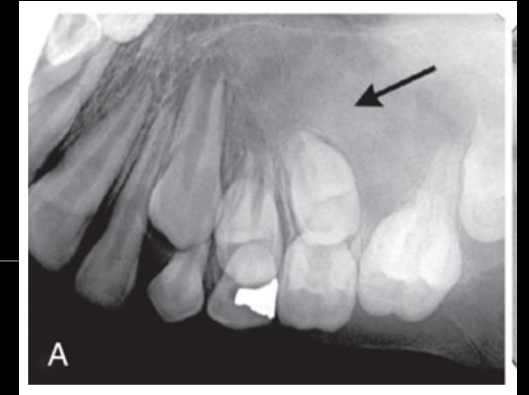
- Hyperparathyroidism (bilateral, no enlargement), Paget disease (cotton-wool, bilateral).

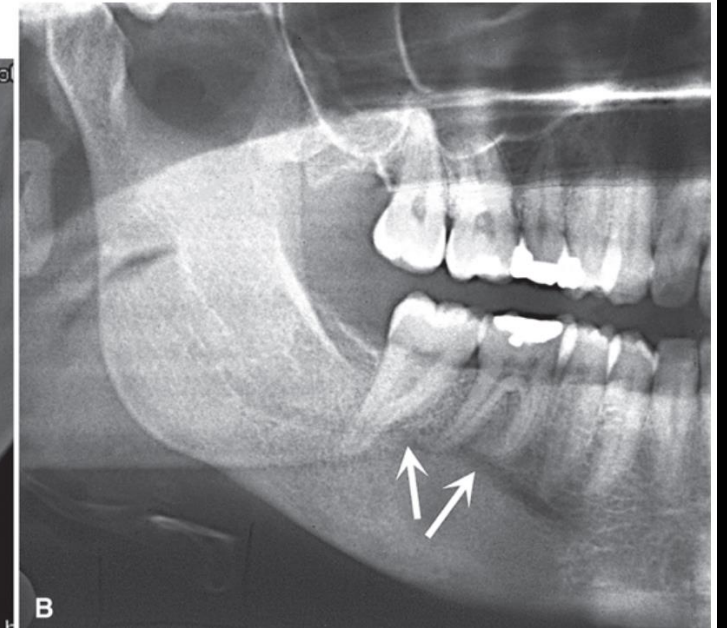
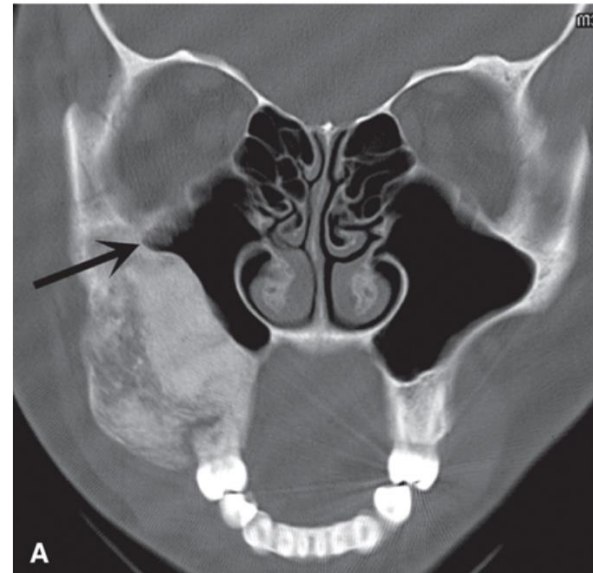
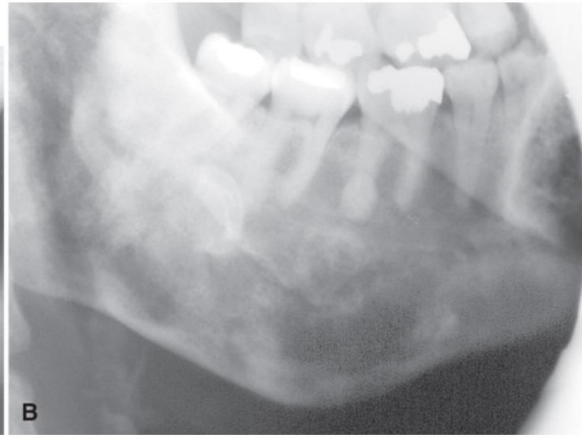
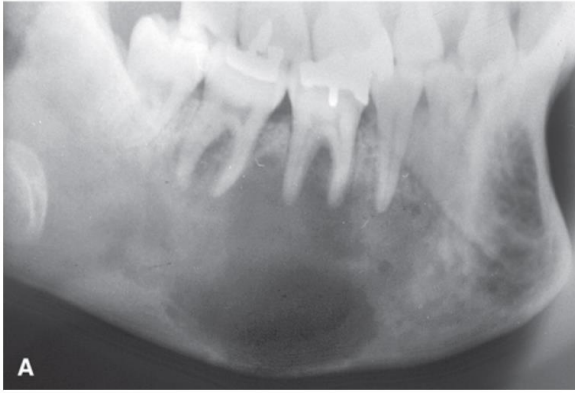
Key Differentials:

- Osteomyelitis (periosteal bone, sequestra),
- Osteosarcoma (aggressive, malignant features),
- Cemento-ossifying fibroma (concentric growth, tooth displacement).

Management:

- Diagnosis via clinical + radiologic features; biopsy usually unnecessary.
- Use CBCT/MDCT for detailed assessment and baseline imaging.
- Monitor until skeletal maturity; delay orthodontic/cosmetic procedures.
- Rare risk of sarcomatous transformation (especially post-radiation).
- Avoid biopsy unless essential (infection risk).





Paget Disease

Mechanism:

- Polyostotic disorder; linked to paramyxovirus and mutations (SQSTM1 most common).
- Abnormal osteoclast activity → excessive resorption → poor-quality woven bone.

Clinical Features:

- Common in UK, Australia; declining globally.
- Affects men > women, especially over age 65.
- Bone enlargement → skull, spine, legs, jaws.
- Dental effects: tooth movement, diastema, poor denture fit.
- Elevated serum alkaline phosphatase and urinary hydroxyproline.

Paget Disease

Imaging Features:

- **Location:** Pelvis, femur, skull, vertebrae; maxilla > mandible.

Stages:

- Early: radiolucent.
- Intermediate: ground-glass.
- Late: cotton-wool radiopacity.

Effects on Jaws: Enlarged bone, laminated cortex, sinus floor involvement.

Teeth: Lamina dura loss, irregular hypercementosis.



Paget Disease

Differential Diagnosis:

- **Fibrous Dysplasia:** Younger age, unilateral, sinus encroachment.
- **PCOD:** Focal, older age, periapical origin, radiolucent rim.
- **Metabolic Bone Disease:** Bilateral, no bone enlargement.
- **Cemento-Ossifying Fibroma:** Concentric growth, tooth displacement.
- **Osteomyelitis:** Periosteal bone, sequestra.
- **Osteosarcoma:** Aggressive, malignant features.

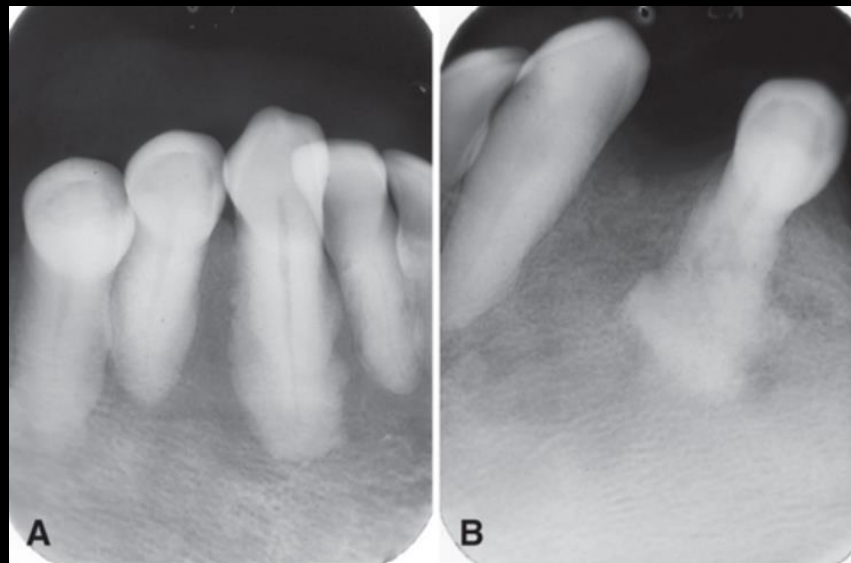
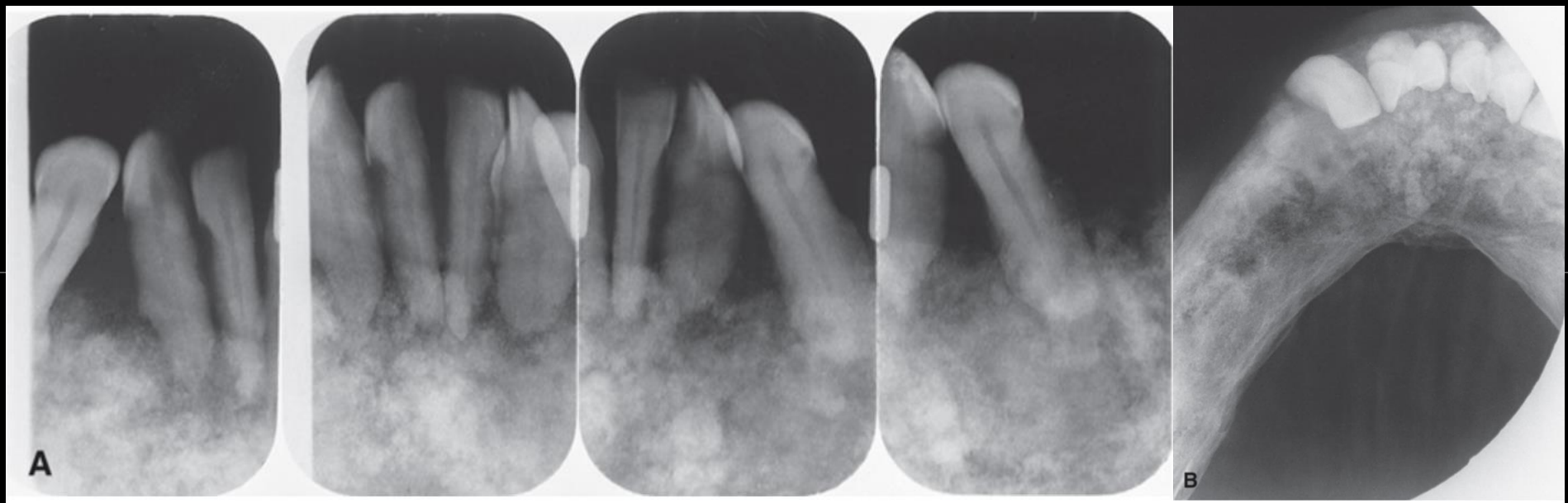
Management:

- **Medical:** Bisphosphonates or calcitonin to reduce bone turnover and pain.
- **Surgical:** For fractures or deformities.

Complications:

- Slow healing post-extraction.
- Increased risk of osteomyelitis.
- ~10% risk of osteosarcoma in polyostotic cases.





Hemoglobinopathies

Disease Mechanism:

- **Sickle Cell Anemia:** HBB gene mutation → sickled RBCs → impaired oxygen delivery.
- **Thalassemia:** HBB or HBA gene mutation → reduced hemoglobin synthesis → anemia.
- Bone marrow hyperplasia occurs as compensation.

Clinical Features:

- **Sickle Cell:** Crises with pain, fever, fatigue, shortness of breath; affects mostly children/adolescents.
- **Thalassemia:** Fatigue, pale skin, “rodent-like” facial features.
- Heart enlargement and murmurs may occur in both.

Hemoglobinopathies

Imaging Features:

- **Jaws:** Osteopenia, enlarged marrow spaces, coarse trabeculae, sinus underdevelopment.
- **Skeleton:** Thinned cortices, widened diploic space, “hair-on-end” skull pattern.
- Bone infarctions and osteomyelitis may occur.

Management:

- **Sickle Cell:** Supportive care; bone marrow transplant (limited use).
- **Thalassemia Minor:** Iron supplements if deficient.
- **Thalassemia Major:** Regular transfusions + iron chelation.

